

DaCENA

Personalized Exploration of Knowledge Graphs Within a Context

Federico Bianchi, Matteo Palmonari, Flavio De Paoli and Marco Cremaschi



ITIS Lab – Innovative Technologies for Interaction and Services



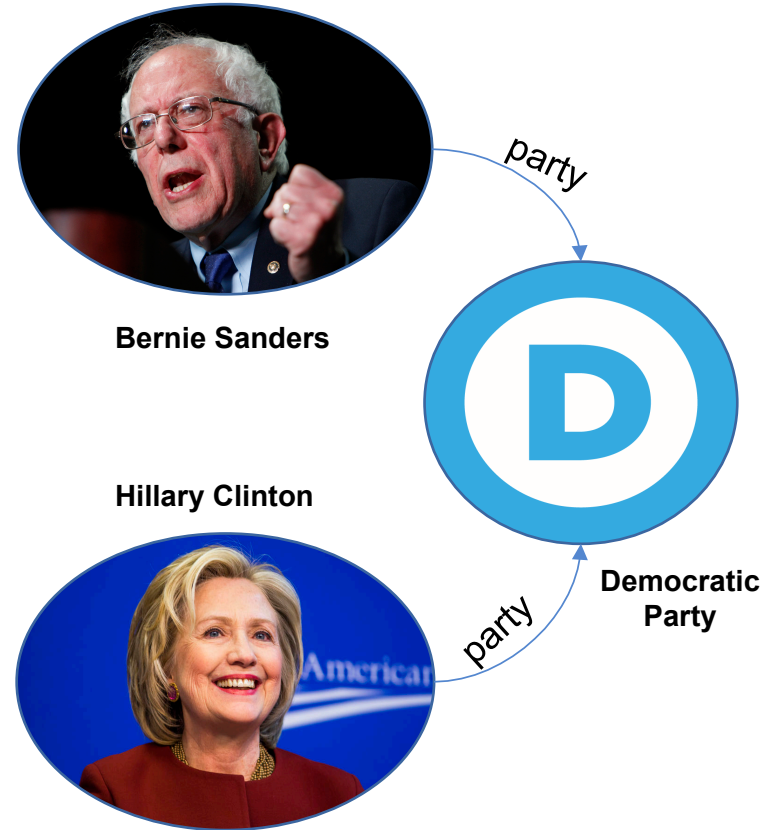
*Dipartimento di Informatica, Sistemistica e Comunicazione
Università degli Studi di Milano-Bicocca*

Outline

- Contextual Exploration of Knowledge Graphs & DaCENA
- Actively Learning to Rank Semantic Associations
- Experiments
- Conclusions and Future Work
- More about DaCENA

Knowledge Graphs (KGs)

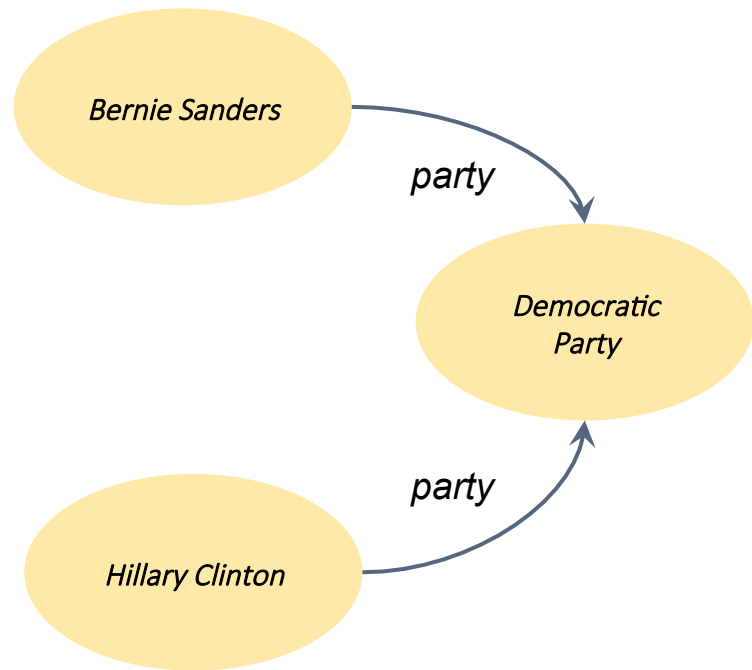
- Models used for knowledge representation using graphs
- DBpedia, YAGO, Google KG, ...
- Nodes represent real-world **entities**
- Labelled edges represent **relations** between them.



KGs may contain interesting relations for users

Relational Knowledge in KGs and Semantic Associations

- KGs provide vast amount of relational knowledge
- Semantic Associations (SAs)
 - chains of relations between entities
 - arbitrary length
 - inverse properties included



Bernie_Sanders party > Democratic_Party < party Hilary_Clinton

Empowering Comprehension of Web Content

Bernie Sanders: I will work with Hillary Clinton to stop Donald Trump

In a live-streamed speech, the Vermont senator made it clear he is no longer actively challenging Clinton and that the goal is to 'defeat Trump badly'

Dan Roberts in Washington 

Friday 17 June 2016 02:08 BST

Bernie Sanders has urged his supporters to look beyond the Democratic presidential nomination in a speech that stopped short of fully endorsing Hillary Clinton but made clear he was no longer actively challenging her candidacy.

In an anticlimatic speech that signalled the effective end of a 14-month campaign odyssey, the Vermont senator insisted his “political revolution continues” despite Clinton’s effective victory in the delegate race.

Who is Bernie Sanders? What is his relation with Hillary Clinton?



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Who is Bernie Sanders? What is his relation with Hillary Clinton?

Bernie Sanders

From Wikipedia, the free encyclopedia

Bernard Sanders (born September 8, 1941), better known as **Bernie Sanders**, is an American politician who has been the [junior United States Senator](#) from [Vermont](#) since 2007. Sanders is the longest serving [independent](#) in U.S. congressional history. Since his election to the [House of Representatives](#) in 1991, he has caucused with the [Democratic Party](#), which has entitled him to [congressional committee](#) assignments and at times given Democrats a majority. Sanders

Bernie Sanders



Contextual Exploration of KGs

Support a user who is doing a familiar task, e.g., *reading a news article*, to access content extracted from a KG, selected and pushed to him in a proactive fashion.

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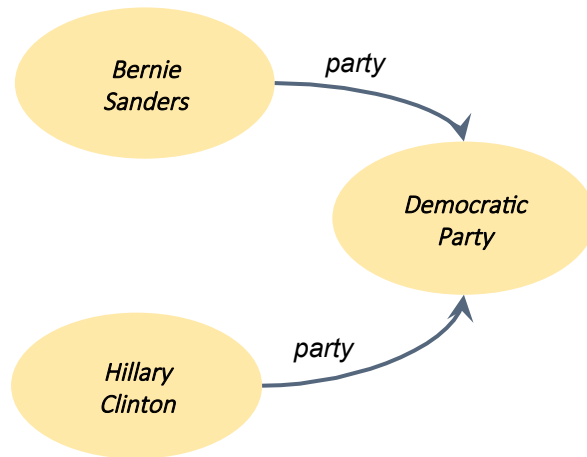
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Contextual Exploration of DBpedia with DaCENA

BACK

A Threat to Spanish Democracy

By CAYETANA ÁLVAREZ DE TOLEDO, NÚRIA AMAT and MARIO VARGAS LLOSA

NOV. 7, 2014

Original Article

MAIN ENTITY: **CATALONIA**

On Sunday, in defiance of an order from the Spanish Constitutional Court, a farcical referendum on Catalan **separatism** will take place. It is not clear who is holding it, as there are neither official voters' rolls nor auditors, only a semiofficial cabal of volunteers. All this in the name of "democracy" and the right to decide, but without respect for the rule of law or the true will of the people, and without acknowledging the gravity of

Serendipity ?

Threshold ?

Path length

relevant

unexpected

100 / 40107

1 2 3

SHOW THE COMPLETE NETWORK

Part of the graph extracted from DBpedia

Catalonia	< isPartOf	Riudaura	leaderParty	>	Republican Left of Catalonia	ideology	>	Separatism
Catalonia	< isPartOf	Vilallonga del Camp	leaderParty	>	Republican Left of Catalonia	ideology	>	Separatism
Catalonia	< isPartOf	Pardines, Girona	leaderParty	>	Republican Left of Catalonia	ideology	>	Separatism
Catalonia	< isPartOf	Alcanar	leaderParty	>	Republican Left of Catalonia	ideology	>	Separatism
Catalonia	< isPartOf	Sant Feliu Sasserra	leaderParty	>	Republican Left of Catalonia	ideology	>	Separatism
Catalonia	< isPartOf	Cabrera d'Anoia	leaderParty	>	Republican Left of Catalonia	ideology	>	Separatism
Catalonia	< isPartOf	Almoster	governmentType	>	Republican Left of Catalonia	ideology	>	Separatism
Catalonia	< isPartOf	Barcelona	headquarter	>	Republican Left of Catalonia	ideology	>	Separatism
Catalonia	< birthPlace	Juan Carretero i Grau	party	>	Republican Left of Catalonia	ideology	>	Separatism

www.dacena.org

(Palmonari&al, 2015)

Entity Extraction

Bernie Sanders has urged his supporters to look beyond the Democratic presidential nomination in a speech that stopped short of fully endorsing Hillary Clinton but made clear he was no longer actively challenging her candidacy. In an anticlimatic speech that signalled the effective end of a 14-month campaign odyssey, the Vermont senator insisted his “political revolution continues” despite Clinton’s effective victory in the delegate race.

Entities are extracted from text



Entity Extraction

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Entities are extracted from text

Bernie Sanders, Hillary Clinton,
Democratic Party, Vermont...

Entity Extraction



SAs Retrieval

Retrieval of Semantic Associations

SPARQL query to the DBpedia endpoint.

SAs between all the entities

- maximum number of hops = 2



Retrieval of Semantic Associations

SPARQL query to the DBpedia endpoint.

SAs between all the entities

- maximum number of hops = 2

Between (Bernie Sanders **and** Hillary Clinton)



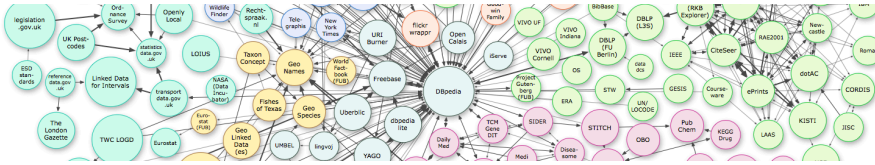
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Entity Extraction



SAs Retrieval

Retrieval of Semantic Associations

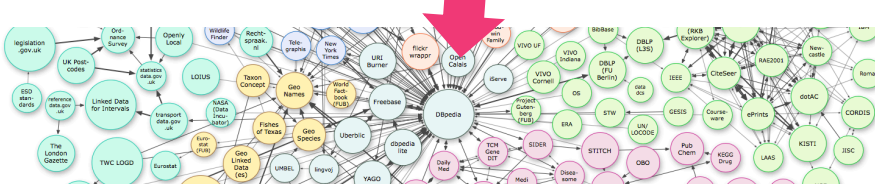
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SPARQL



Entity Extraction



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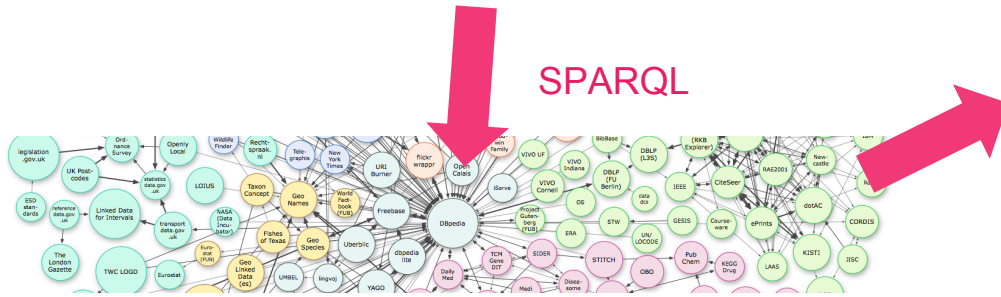
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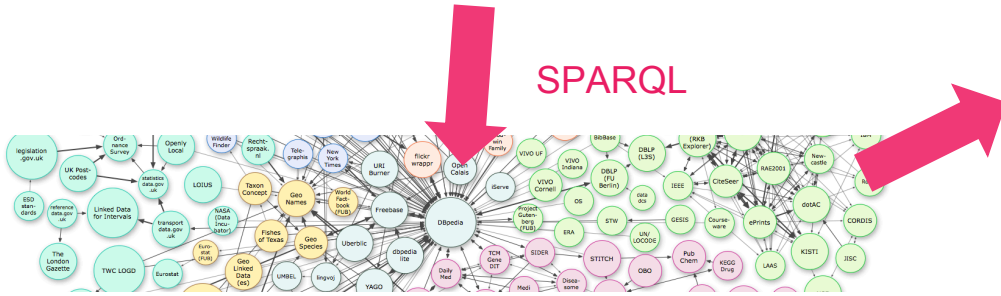
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Between (Bernie Sanders and Hillary Clinton)

SPARQL



Bernie Sanders

party

*Democratic
Party*

Entity Extraction



SAs Retrieval

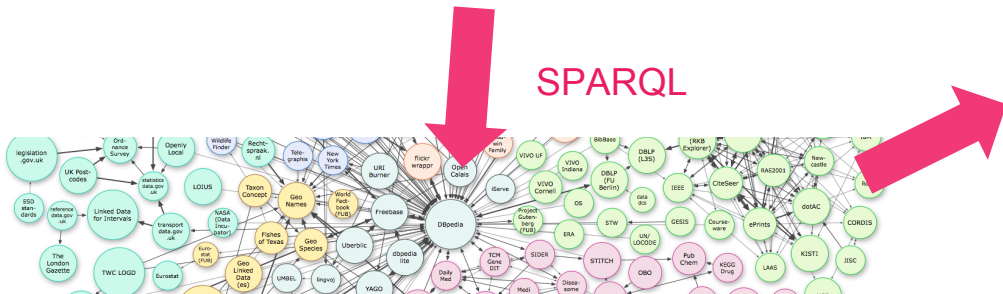
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SPARQL



Bernie Sanders

party

*Democratic
Party*

Hillary Clinton

party

Entity Extraction

SAs Retrieval

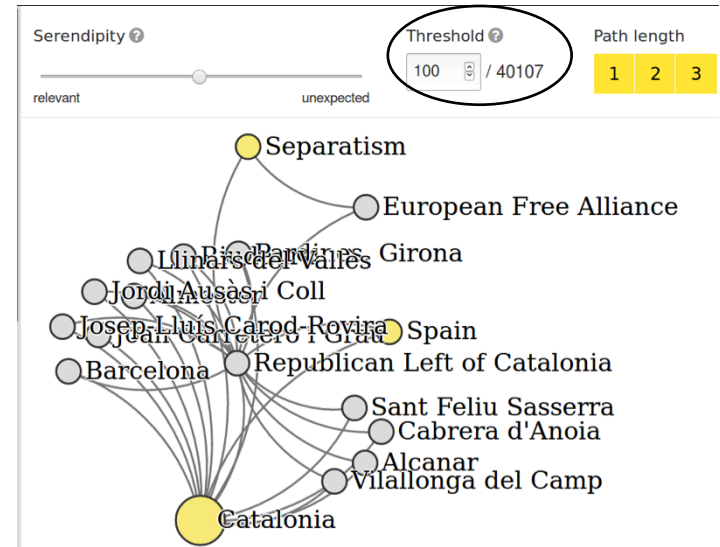
Information Overload in Contextual KG Exploration

Too many associations from even small pieces of text

- *E.g.*, 40107 associations from an article with 942 words
- Not fit in a single screen
- Users can explore only a limited number of associations (≤ 100)

Crucial issue for KG exploration:

Which are the most interesting to show to users?



Ranking SAs by Estimated Interest: Serendipity

Heuristic measure: try to find those associations that are relevant and may be unexpected to users

Serendipity = relevance + unexpectedness

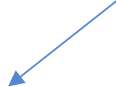
$$\text{Serendipity}(\text{SA}, \text{Text}) = \alpha * \text{relevance}(\text{SA}, \text{Text}) + (1 - \alpha) * \text{rarity}(\text{SA})$$

Ranking SAs by Estimated Interest: Serendipity

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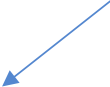

relevance (SA, Text) = $\cos(\textit{abstracts}(\textit{SAs}), \textit{Text})$
- with TF-IDF weighting

Ranking SAs by Estimated Interest: Serendipity

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 $\text{relevance}(\text{SA}, \text{Text}) = \cos(\text{abstracts}(\text{SAs}), \text{Text})$ (Aleman-Meza&al, 2005)
- with TF-IDF weighting

 Inverse of frequency normalized over total frequency

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user can tune the weight assigned to relevance vs unexpectedness

BACK

Serendipity ?

relevant ————— unexpected

Threshold ? 100 / 40107

Path length 1 2 3

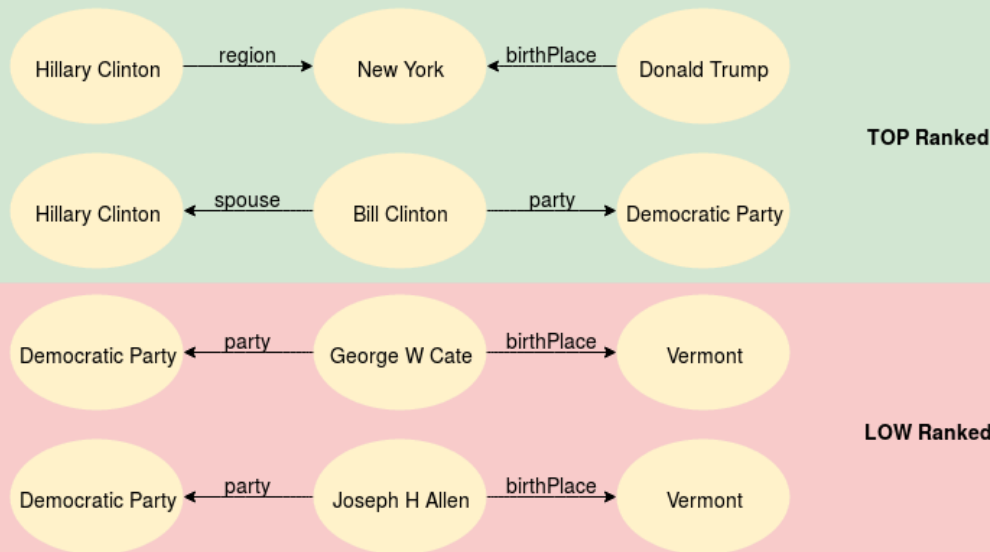
SHOW THE COMPLETE NETWORK

Separatism

Example of SAs ranked by Serendipity

Article Text

Bernie Sanders has urged his supporters to look beyond the **Democratic** presidential nomination in a speech that stopped short of fully endorsing **Hillary Clinton** but made clear he was no longer actively challenging her candidacy. In an anticlimatic speech that signalled the effective end of a 14-month campaign odyssey, the **Vermont** senator insisted his “political revolution continues” despite Clinton’s effective victory in the delegate race. But crucially, he implied he would soon be working with her campaign to help defeat **Donald Trump**. “The major political task that we face in the next five months is to make certain that Donald Trump is defeated and defeated badly,” Sanders told supporters in a live-stream video. “And I personally intend to begin my role in that process in a very short period of time.”



Outline

- Contextual Exploration of Knowledge Graphs
- **Actively Learning to Rank Semantic Associations**
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Personalized Exploration of KGs

What if different users are interested in **different** SAs?

1. Learn a ranking function starting from **explicit ratings** given by the users
2. Ask users **few ratings as possible** (rating too many SAs can become a tedious task)
3. Speed up learning by **sampling** SAs that are estimated more informative for training the model

Definition of an active learning to rank model for personalized contextual exploration of KGs

Active Learning to Rank for SAs



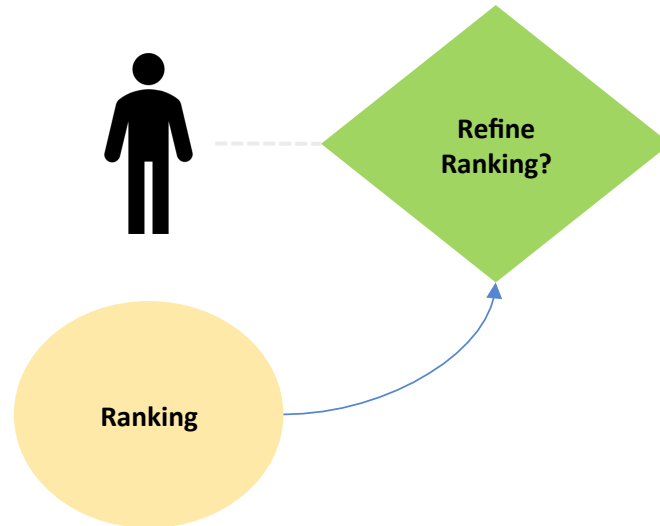
Active Learning to Rank for SAs

Rank SVM algorithm:

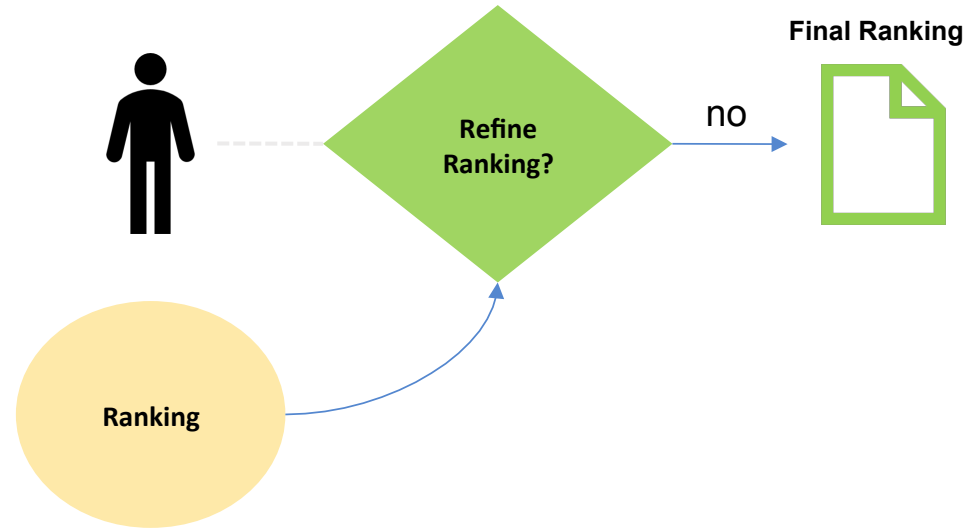
- Derivated From SVM (Support Vector Machine)
- Well known and widly used in the literature



Active Learning to Rank for SAs



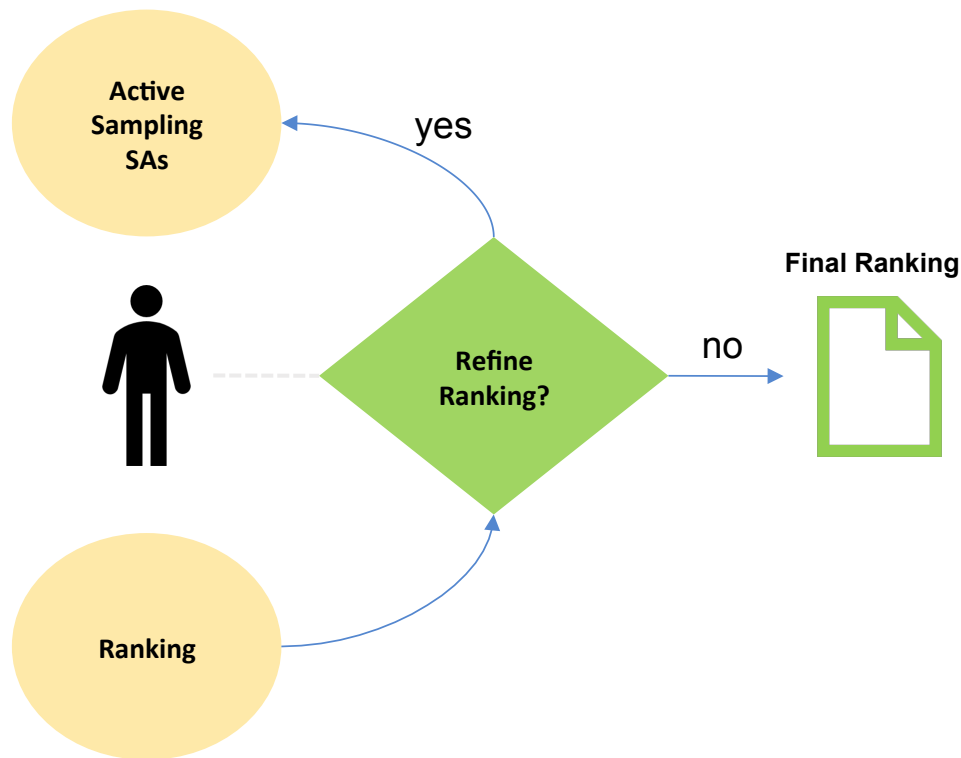
Active Learning to Rank for SAs



Active Learning to Rank for SAs

Two algorithm used to find meaningful SAs:

- Pairwise Sampling (**PS**)
(Qian&al, 2013)
- AUC-Based Sampling (**AS**)
(Donmez&al, 2009)

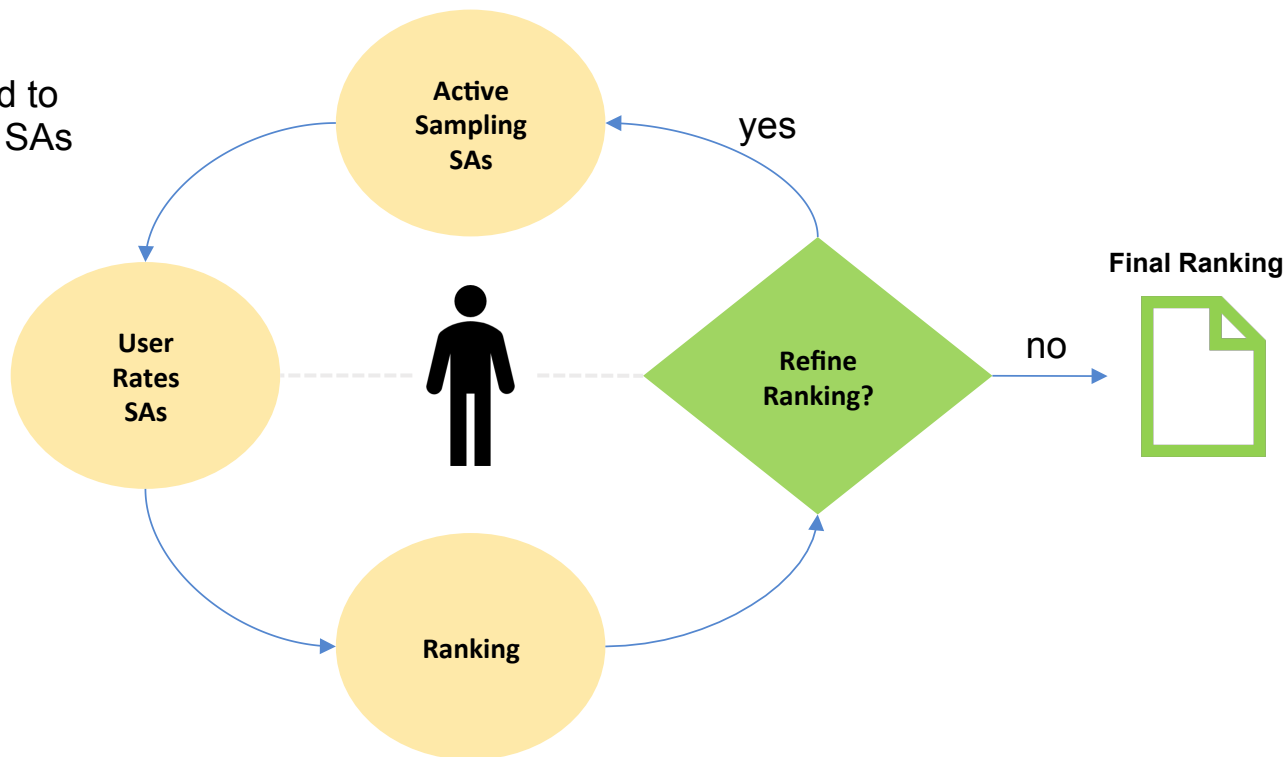


Active Learning to Rank for SAs

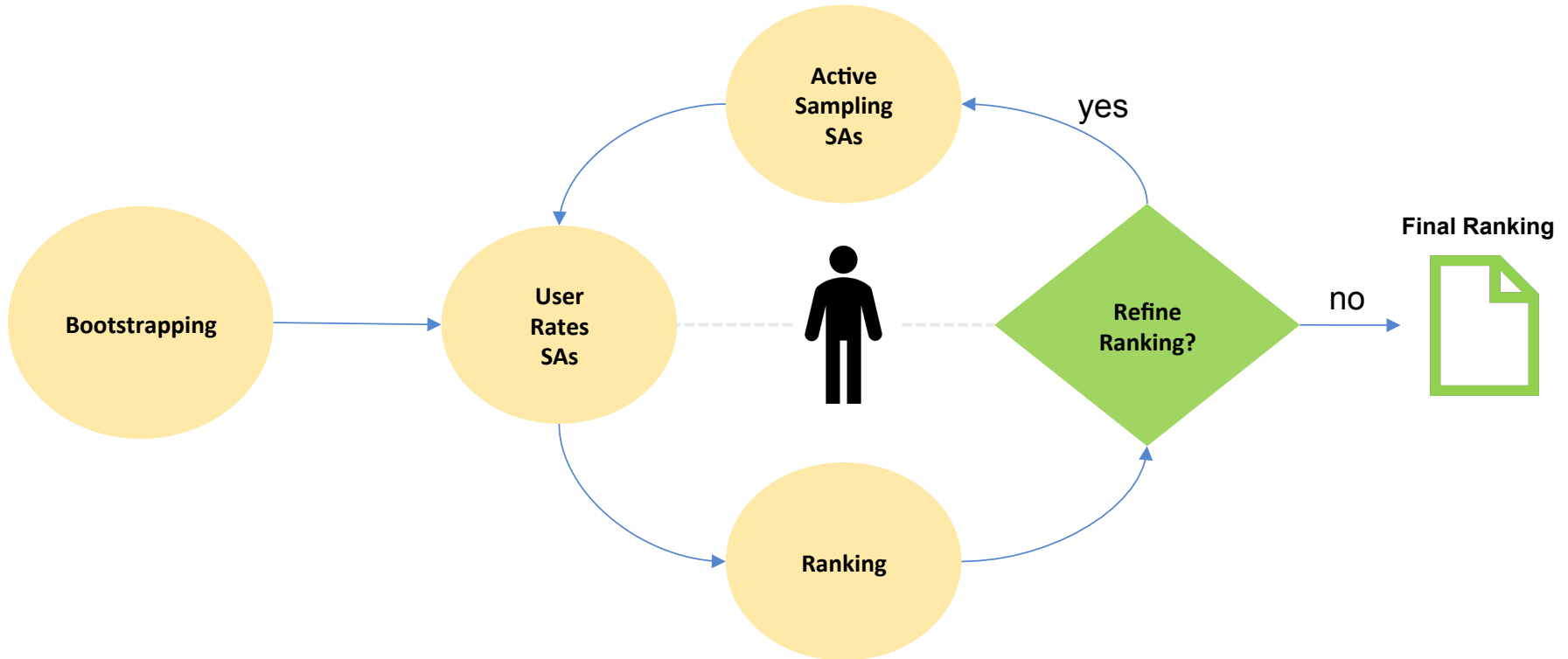


Active Learning to Rank for SAs

But, ranking models need to be initialized with ranked SAs (cold-start problem)

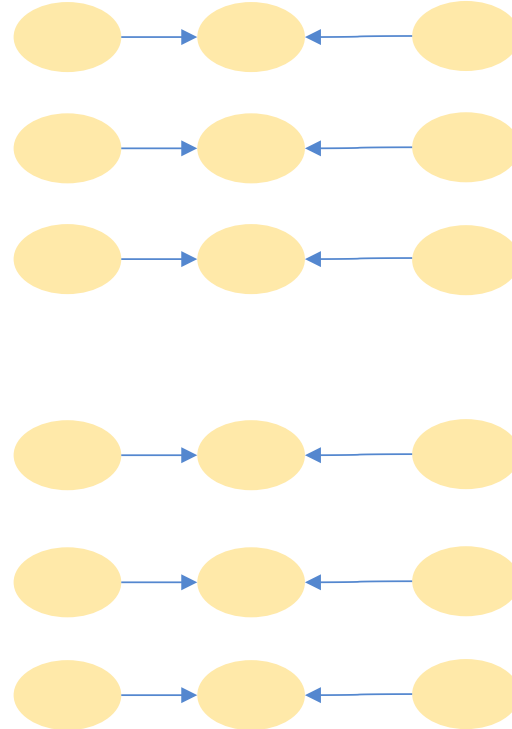


Active Learning to Rank for SAs



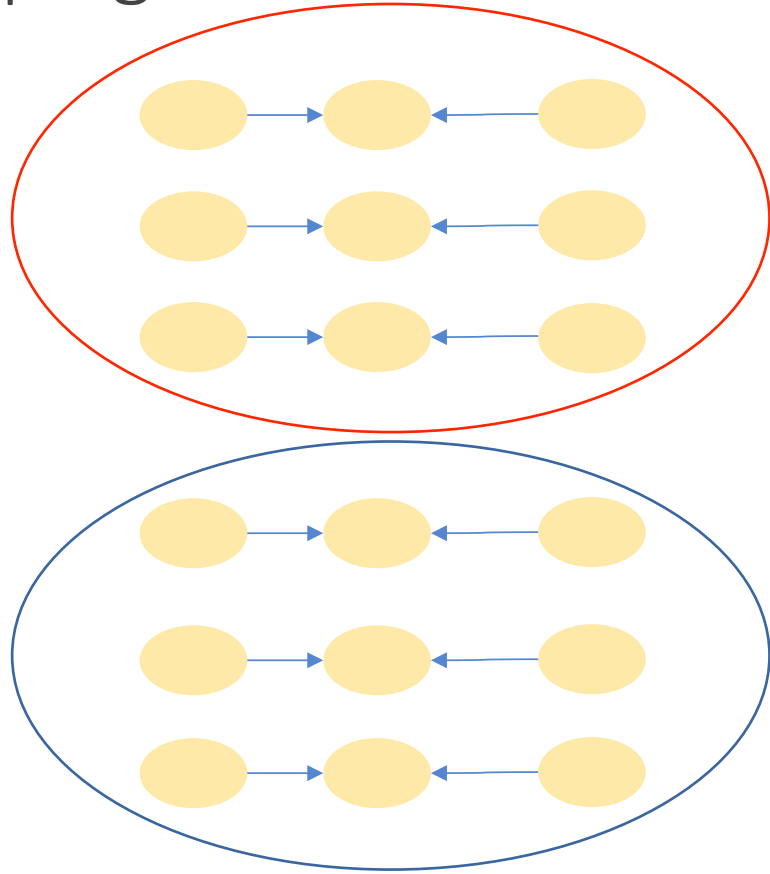
Clustering as Bootstrapping

- Use clustering algorithms on the set of SAs
- For each cluster select the SA that is closest to the cluster average
- User rates all the SAs that represent the clusters



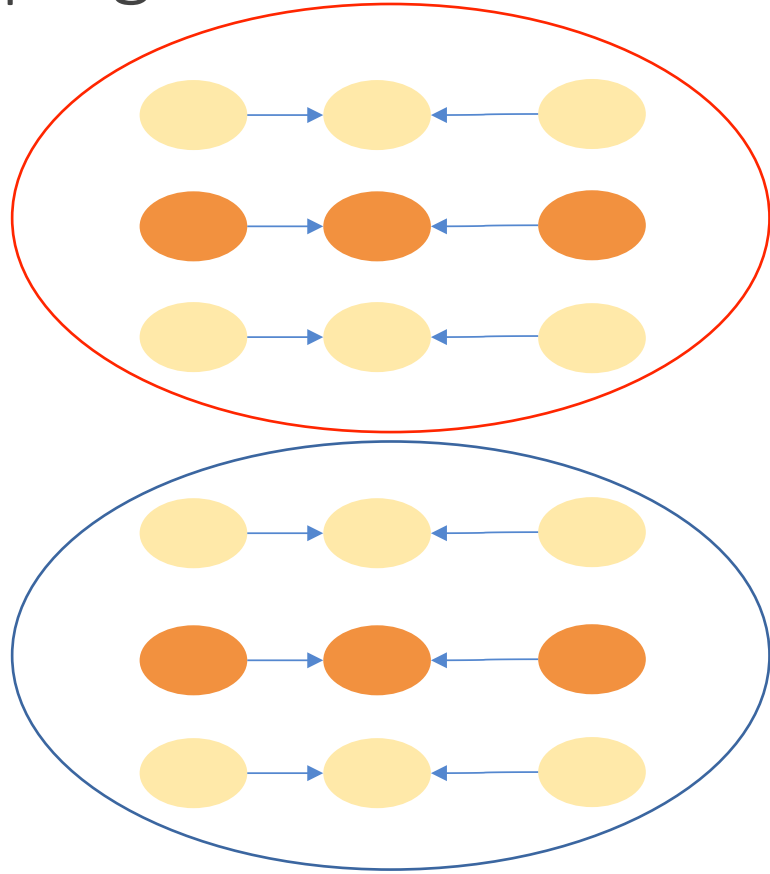
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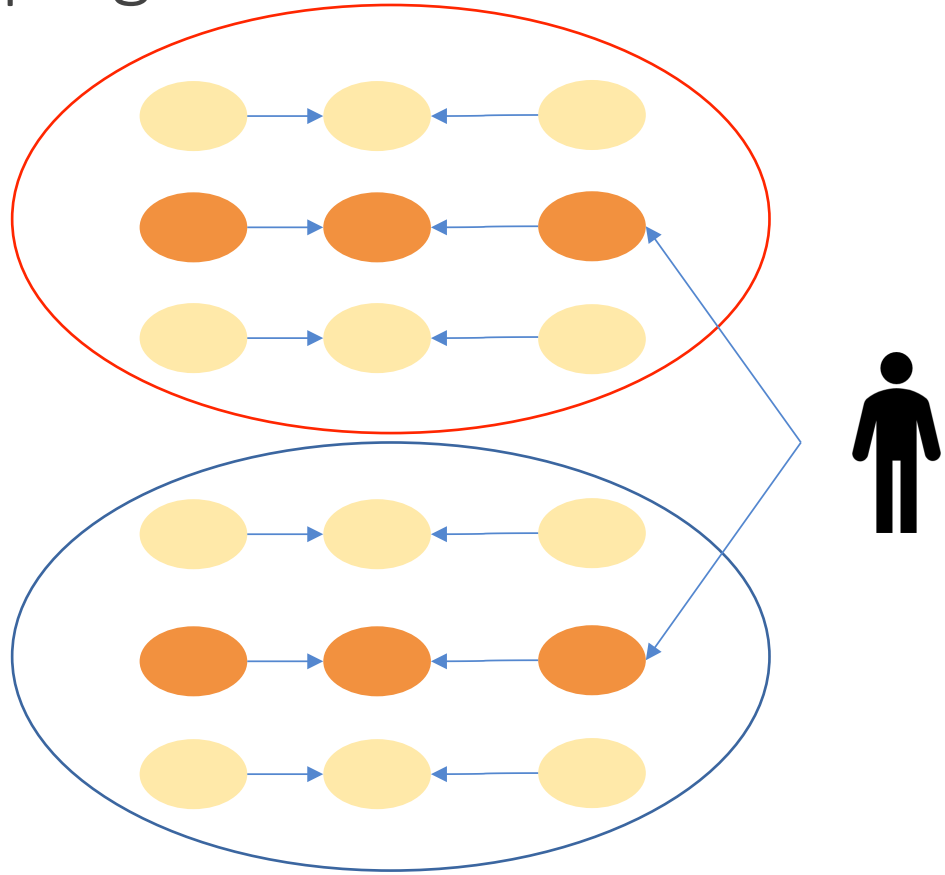
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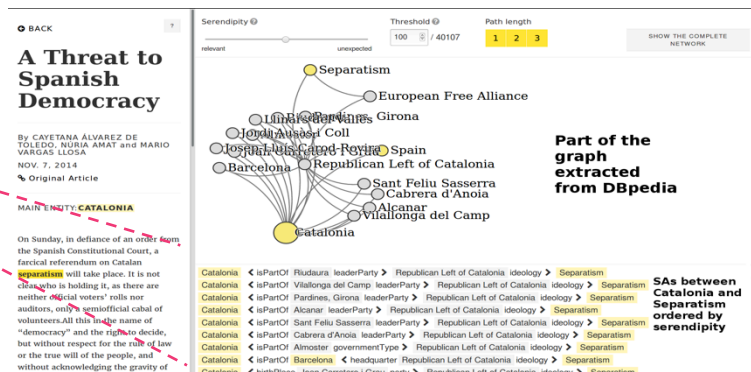
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Serendipity as Bootstrapping

- User rates top-k SAs ranked by Serendipity
- Users are able to see an ordered set of SAs since the beginning
- Users rates SAs that are estimated to be interesting for them



Serendipity vs Clustering

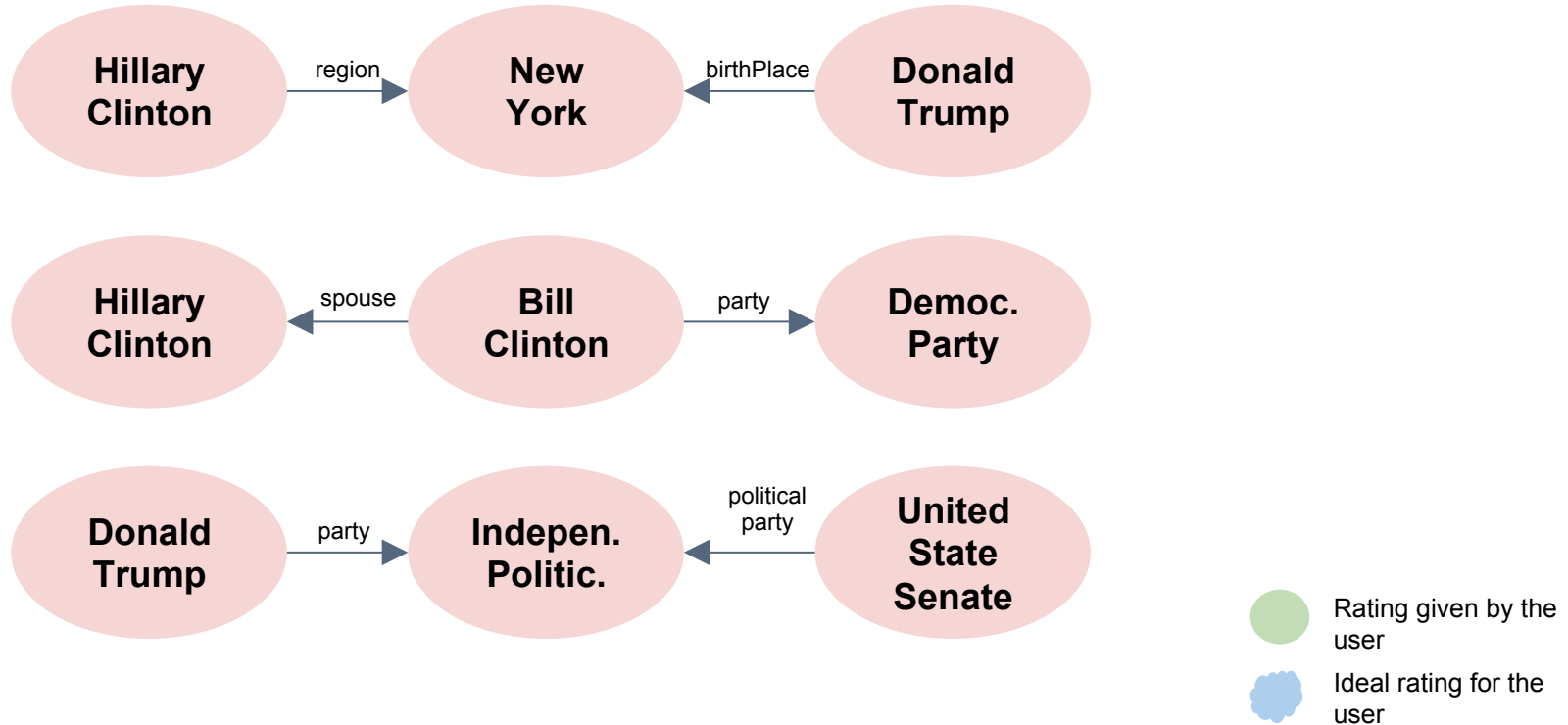
Clustering:

- PROS: selected SAs are representative of the vector space
- CONS: rated SAs might not be interesting for the user

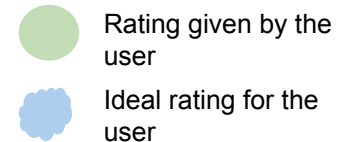
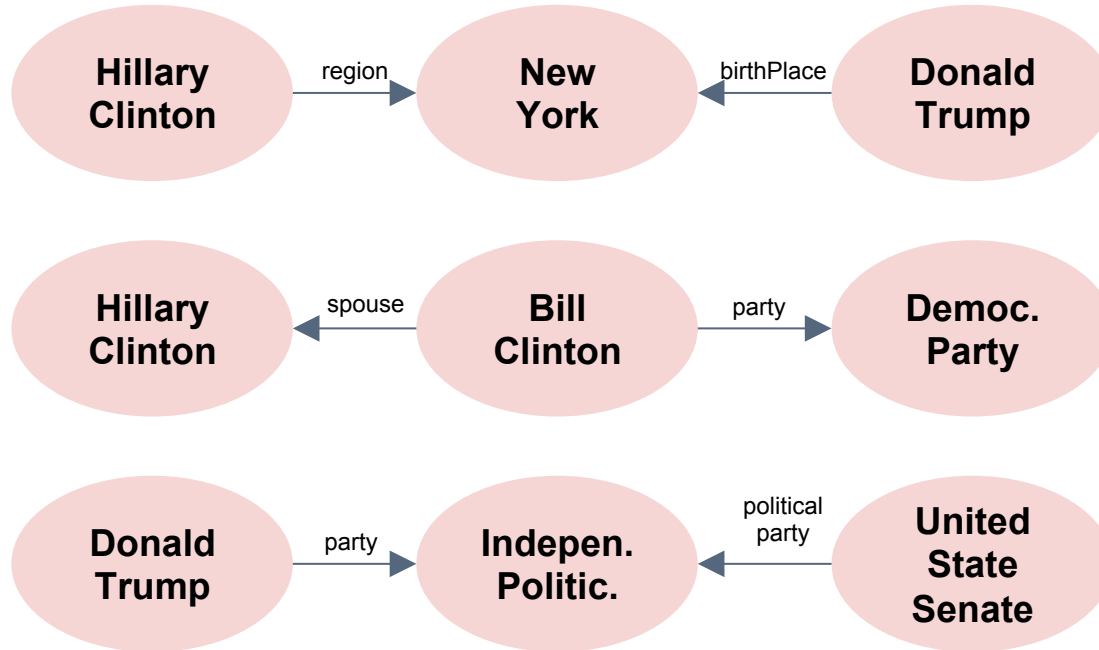
Serendipity:

- PROS: rated SAs are estimated to be interesting for a generic user
- CONS: heuristic function, no representativeness

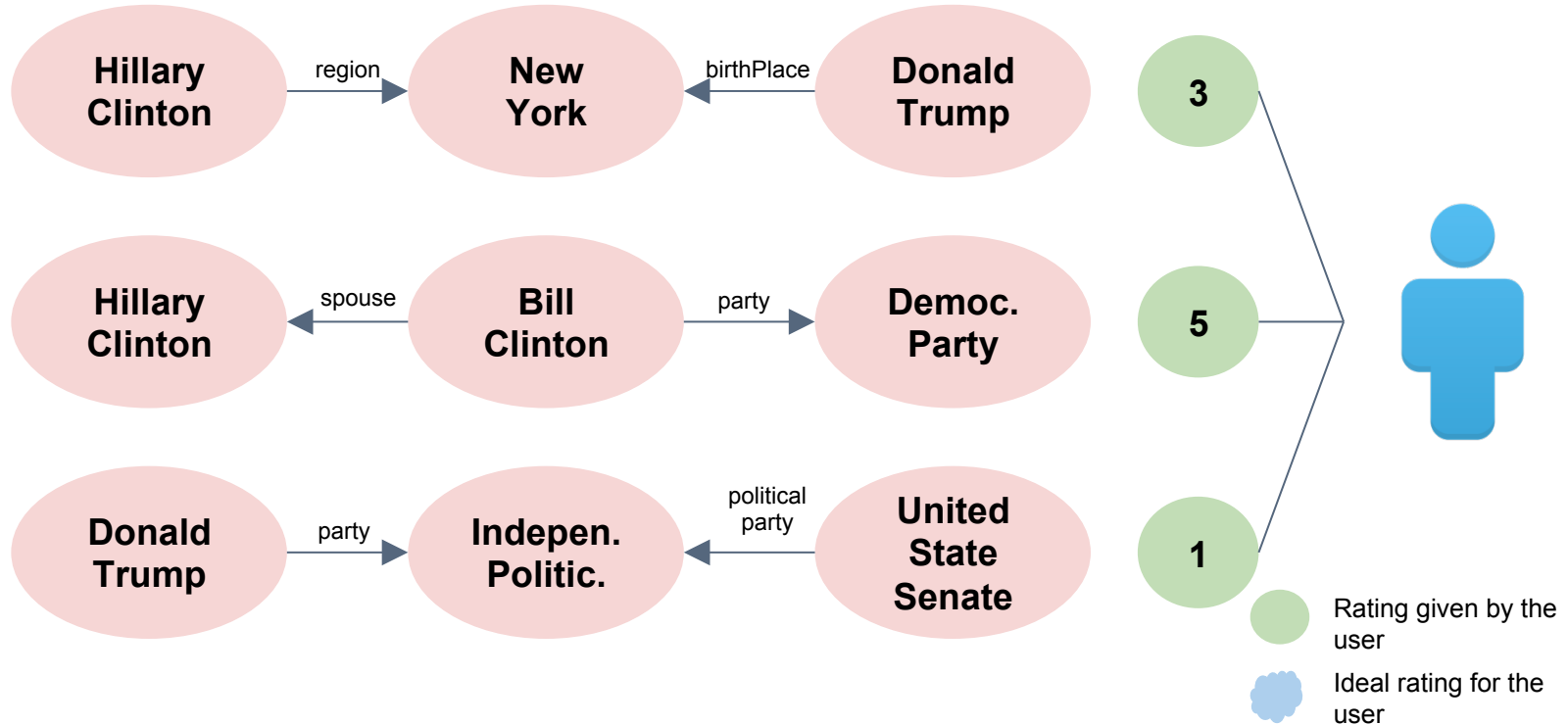
Example: Rating of Most Serendipitous SAs (#0)



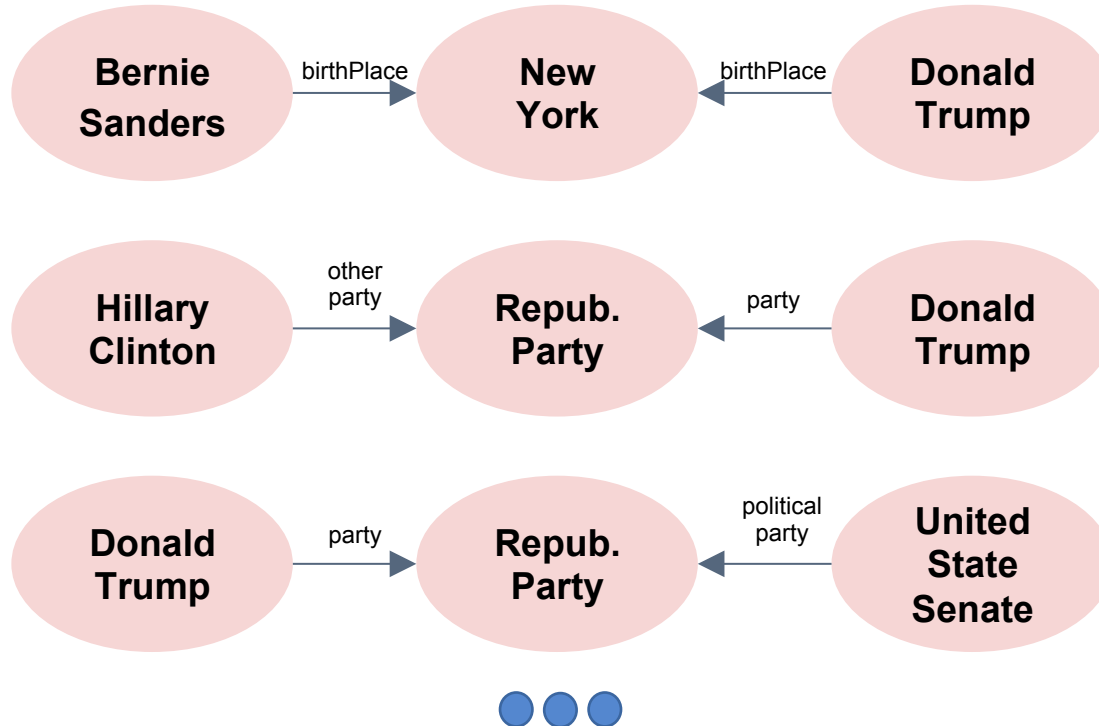
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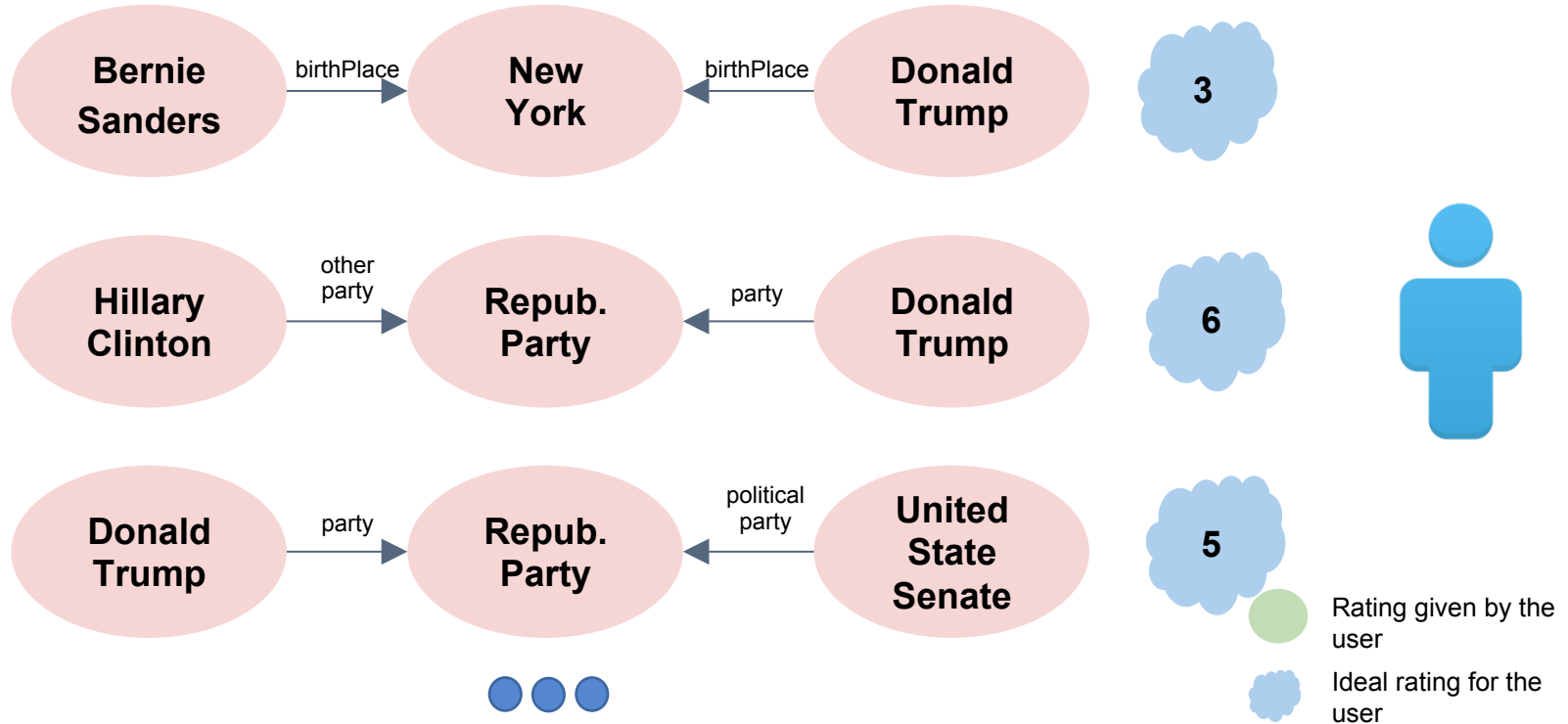
Example: Ranking Learned with RankSVM (#0)



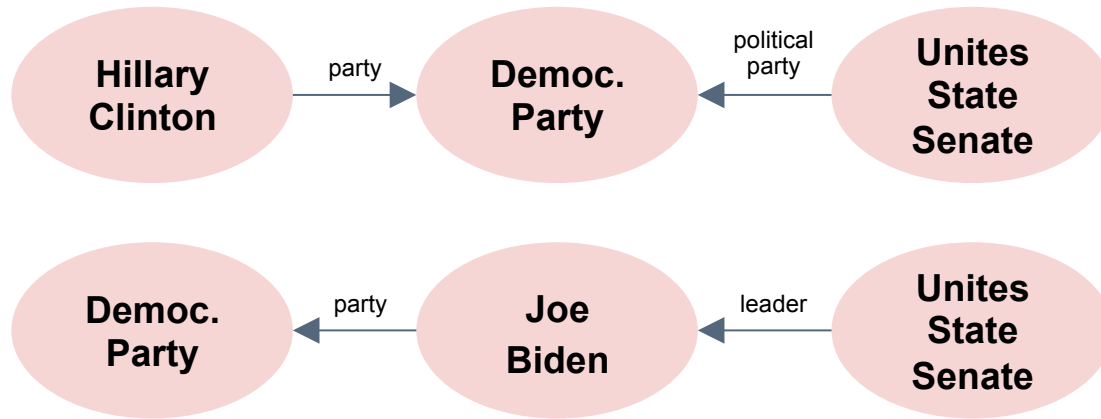
 Rating given by the user

 Ideal rating for the user

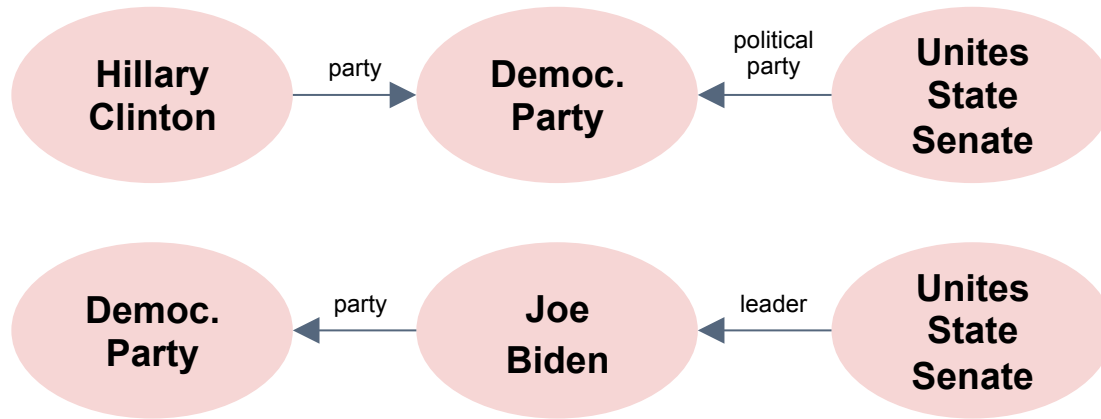
Example: Ranking Learned with RankSVM (#0)



Example: Rating on Sampled SAs (#1)

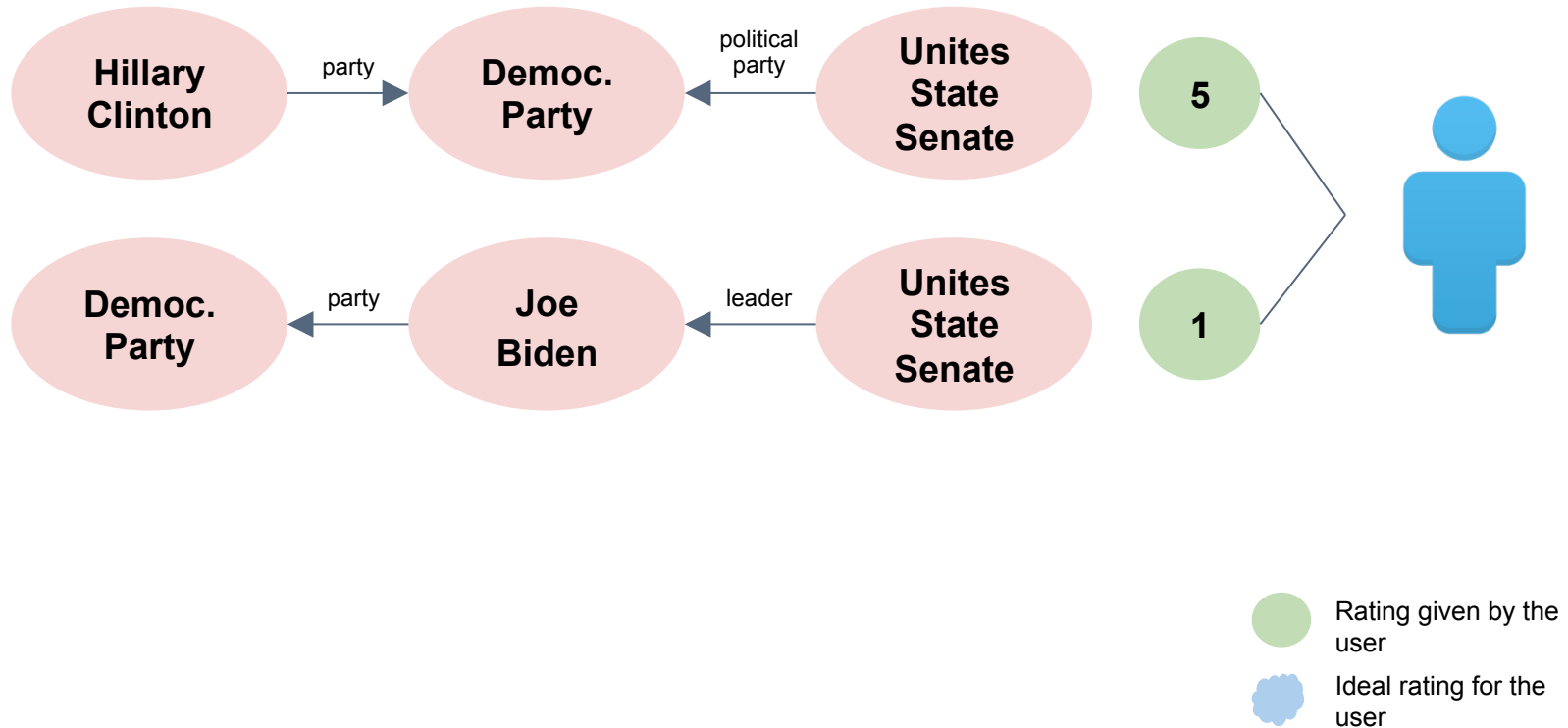


Example: Rating on Sampled SAs (#1)

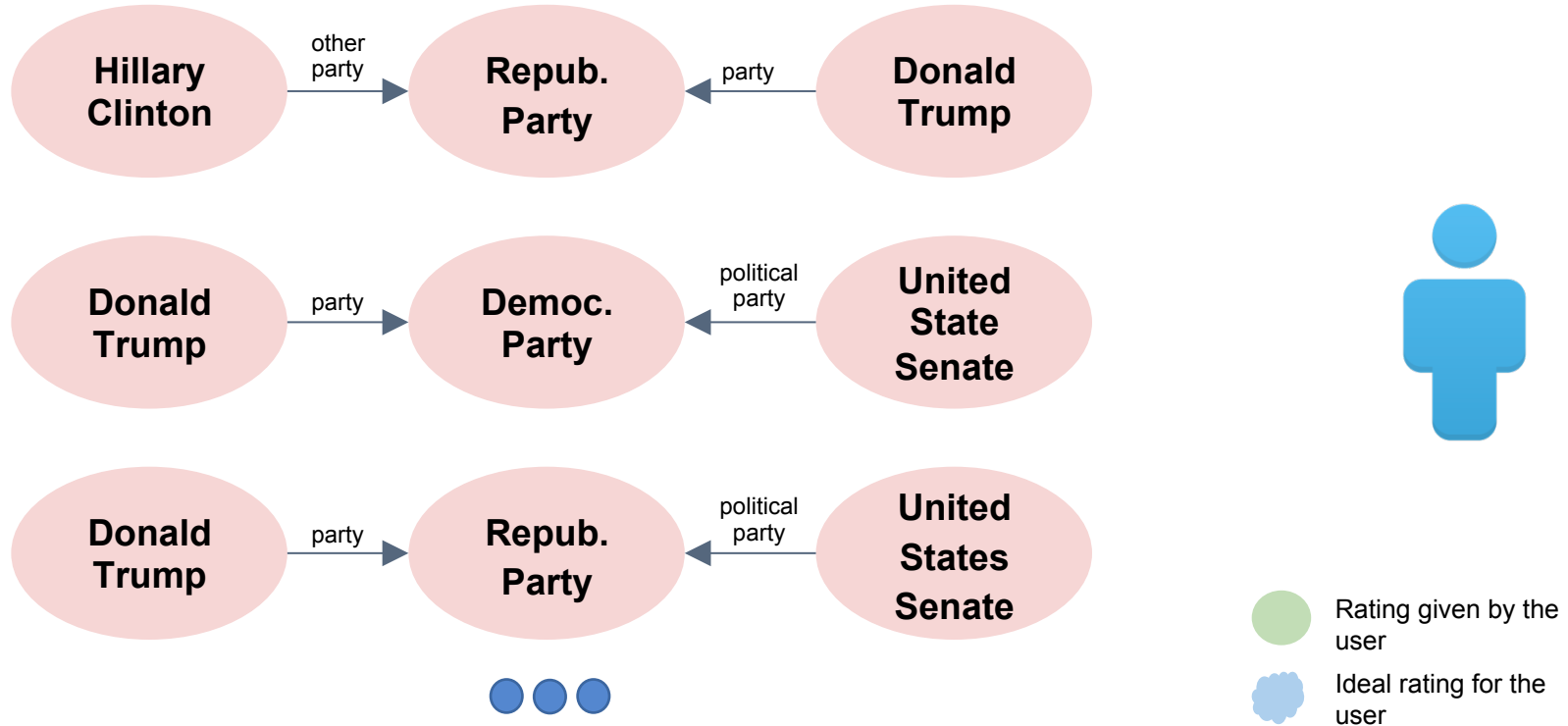


-  Rating given by the user
-  Ideal rating for the user

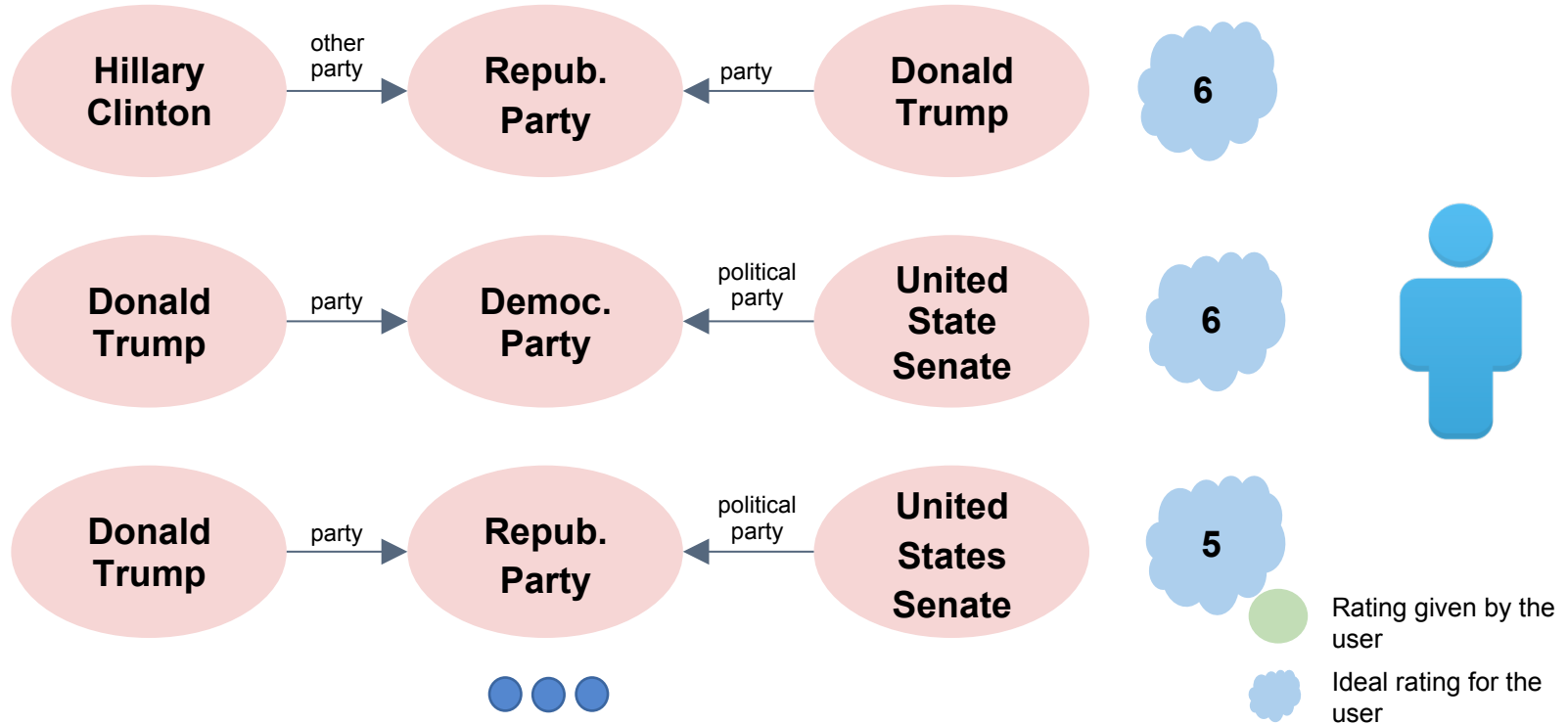
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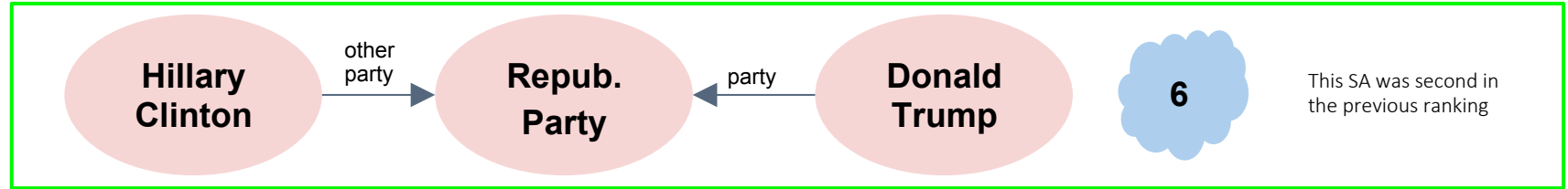
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Example: Ranking Learned with RankSVM (#1)



Rating given by the user

Ideal rating for the user



Features for RankSVM

SAs are represented in the space using different features divided in three main categories:

- **Topological Features:**
 - PageRank on SAs (Page&al, 1999)
 - DBpedia PageRank (Thalhammer&al, 2016)
 - HITS (Kleinberg&al, 1999)
- **Relevance Features:**
 - Relevance (Palmonari&al, 2015),
 - Temporal Relevance (Bianchi&al, 2017)
- **Predicate-Based Features:**
 - Path Informativeness (Pirrò, 2015)
 - Path Pattern Informativeness (Pirrò, 2015)
 - Rarity (Aleman-Meza&al, 2005)

Path Informativeness

Which is the most informative predicate for an entity?

Inspired by **TF-IDF** (Term Frequency - Inverse Document Frequency):

representativeness of a **term** in a **document** is given by both the number of times it appears in the document and the number of times it appears in the whole collection.

A term is present in all the documents it might be less important than others (e.g. “and”, “or”, “it”, “is”...).

Path Informativeness

Path informativeness determines the weight of a predicate p that occurs in a triple (s,p,o)

Relations with TF-IDF:

Here, the term that is weighed is the **predicate** in the triple

Here, a document is a **triple**

(weight of p in a triple (s,p,o) depends on s and o)

Here the collection of documents is the **graph**

“depends on s and o “: when computing the weight of a predicate p in a triple (s,p,o) , we consider all triples (edges) incoming in o AND all triples outgoing from s

Path Informativeness

Given a graph G , the frequency of a predicate p for an entity w is computed using two separate scores:

- A score for edges with p as predicate that are incoming in w
- A score for edges with p as predicate that are outgoing from w

$$\text{pf}_i^w(p, G) = \frac{|\mathcal{E}_i(w)|_{\pi(p)}}{|\mathcal{E}_i(w)|}$$

Numerator: number of triples (edges) **incoming** in entity w with predicate p

Denominator: total number of triples **incoming** in w

$$\text{pf}_o^w(p, G) = \frac{|\mathcal{E}_o(w)|_{\pi(p)}}{|\mathcal{E}_o(w)|}$$

Numerator: number of triples **outgoing** from entity w with predicate p

Denominator: total number of triples **outgoing** from w

Path Informativeness

p is a predicate, G a graph, w is an entity

$$\text{itf}(p, G) = \log \frac{|\mathcal{E}|}{|\mathcal{E}|_{\pi(p)}}$$

Numerator: number of triples

Denominator: number of triples including p

$$\text{pfitf}_x(p, G) = \text{pf}_x \times \text{itf}$$

$x = o$ or i : it means that it can be computed for both incoming and outgoing predicates)

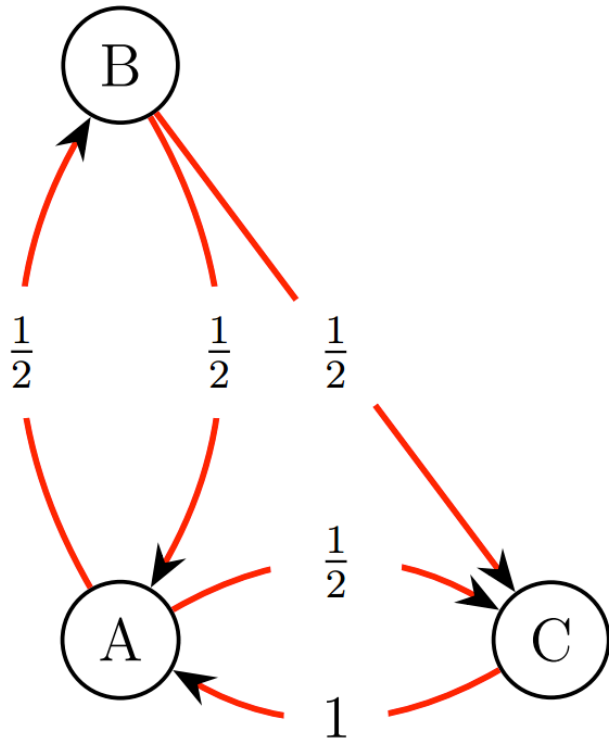
Path Informativeness

Informativeness of a **triple** $(\mathbf{w}_s, p, \mathbf{w}_t)$, is given by average of **outgoing pf-itf** for predicate p and its subject $\mathbf{w}_s$, and **ingoing pf-itf** for predicate p and its object $\mathbf{w}_t$.

If $\mathbf{w}_s$ is the subject and $\mathbf{w}_t$ is the object, then informativeness of a triple $\pi=(\mathbf{w}_s, p, \mathbf{w}_t)$ in G is defined by:

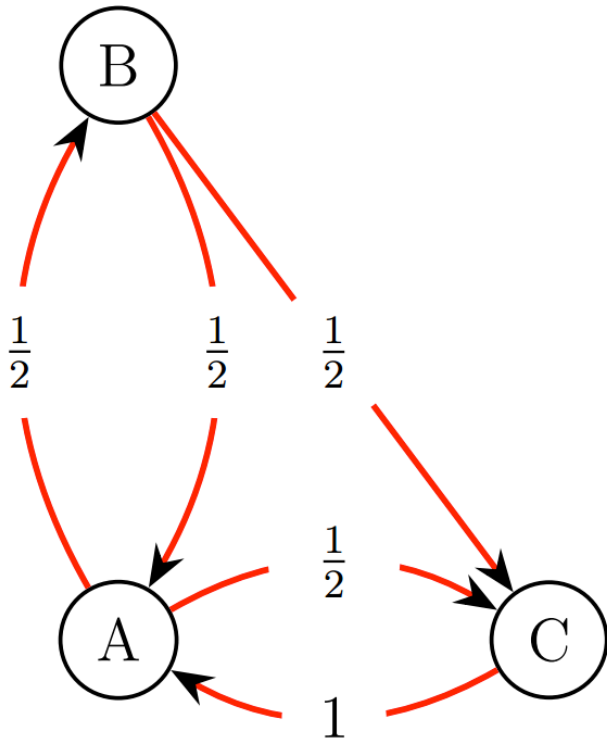
$$I(\pi, G) = [\text{pfitf}_o^{w_s}(p, G) + \text{pfitf}_i^{w_t}(p, G)]/2$$

Which node in a graph is the most important?



How can we understand which node in a graph (entity) is the most important?

PageRank

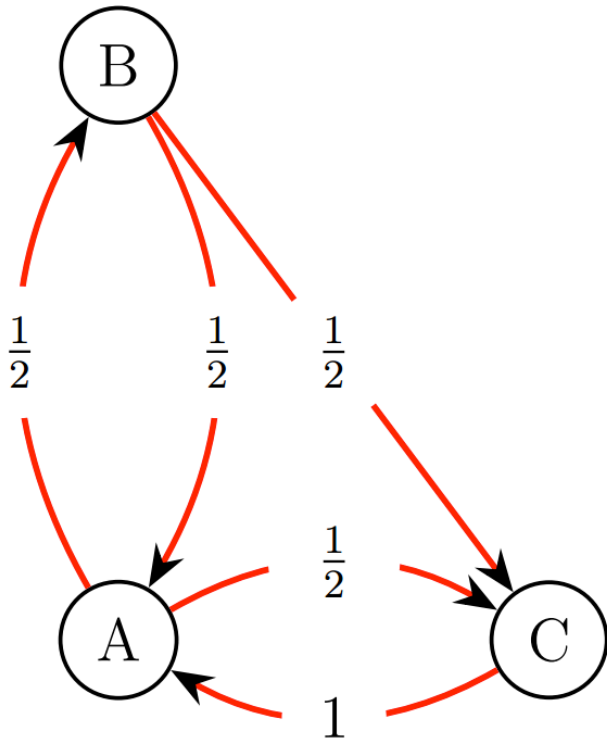


Assumption: A node that points to other nodes distributes equally its *importance* to those other nodes.

This means that if a node has *importance* equal to K and this node points to N nodes, each pointed node receives K/N of importance

Thus: The importance of a **node** is given by the sum of the *importances* it **gets** from the nodes that point to it.

PageRank: equation

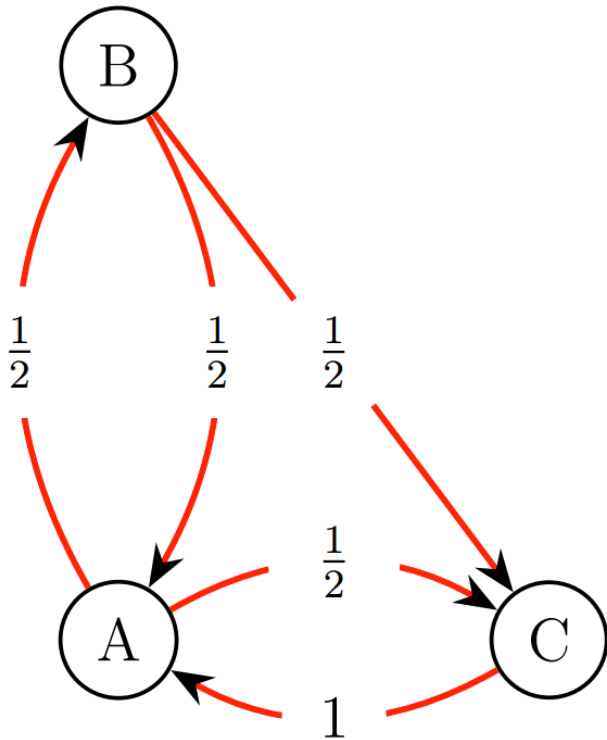


C only points to A, while B points to both A and C. The importance of A is given by the importance of C + half of the importance of B.

Last equation is just a constraint to make everything sum to one.

$$\begin{cases} w_A = w_C + \frac{1}{2}w_B \\ w_B = \frac{1}{2}w_A \\ w_C = \frac{1}{2}w_A + \frac{1}{2}w_B \\ w_A + w_B + w_C = 1 \end{cases}$$

PageRank: result

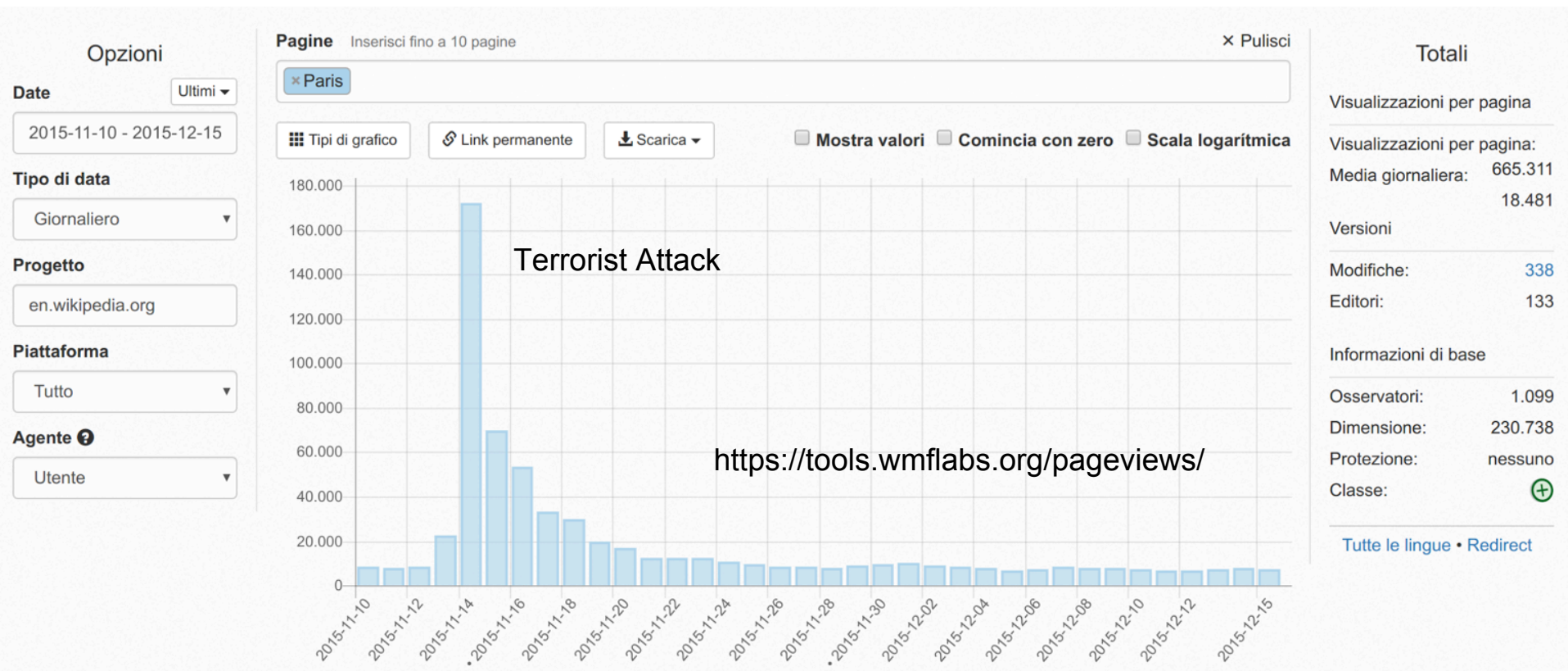


C only points to A, while B points to both A and C. The importance of A is given by the importance of C + half of the importance of B.

Last equation is just a constraint to make everything sum to one.

$$\begin{cases} w_A = \frac{4}{9} = 0.\bar{4} \\ w_B = \frac{2}{9} = 0.\bar{2} \\ w_C = \frac{1}{3} = 0.\bar{3} \end{cases}$$

Temporal Relevance: Paris Pageviews on 11/2015



Temporal Relevance

Idea: we can detect if a wikipedia page and thus an entity in a KG like DBpedia was more important or relevant in a given moment of time. For example, the number of access on the wikipedia page of Paris on 11/2015 suggested that something has happened that day. The temporal relevance of an entity in a given day is given by the number of access that were done on that day on the respective Wikipedia page.

Outline

- Contextual Exploration of Knowledge Graphs
- Actively Learning to Rank Semantic Associations
- **Experiments**
- Conclusions and Future Work

Experiments: Objectives

Validate *personalization hypothesis*

- Are different users interested in different SAs?

Evaluate the performance

- (Quick) improvement of the ranking quality with user ratings with more iterations of feedback
- Comparison of different configurations and baseline algorithms

Experiments: Settings

Gold standards: Ideal rankings collected by asking users to evaluate all SAs extracted from articles or pieces of articles

Evaluation Settings:

Contextual Exploration: ratings on the whole dataset

Cross Validation: ratings on training data, ranking on test data

Measure: quality of generated rankings vs ideal rankings (nDCG)

Experiments: Data

Two different datasets:

LAFU (Large Articles, Few Users)

Complete articles (New York Times)

3 articles, 2 user => 3 ideal ranking

Average number of SAs for article => 2600

Rating from 1 to 3 (1 low interest, 3 high interest)

SAMU (Small Articles, Many Users)

Small pieces of text extracted from articles (New York Times, The Guardian)

5 articles, 14 users => 25 ideal ranking

Average number of SAs for article => 74

Rating from 1 to 6 (1 low interest, 6 high interest. Scale is symmetric)

Experiments: Alternative Configurations and Baselines

Algorithm	Bootstrapping	Active Sampling	Learning
Serendipity AS	Serendipity	AUC-Based Sampling	RankSVM
Serendipity PS	Serendipity	Pairwise Sampling	RankSVM
Dirichlet AS	Dirichlet Clustering	AUC-Based Sampling	RankSVM
Dirichlet PS	Dirichlet Clustering	Pairwise Sampling	RankSVM
Gaussian AS	Gaussian Clustering	AUC-Based Sampling	RankSVM
Gaussian PS	Gaussian Clustering	Pairwise Sampling	RankSVM
Random Random	Random	Random	RankSVM
Random	No Bootstrapping	No Active Learning	No Learning to Rank
Serendipity	No Bootstrapping	No Active Learning	No Learning to Rank

Results: Personalization Hypothesis

Inter Rater Reliability measures to assess the **level of agreement** between users with respect to the same items (SAs).

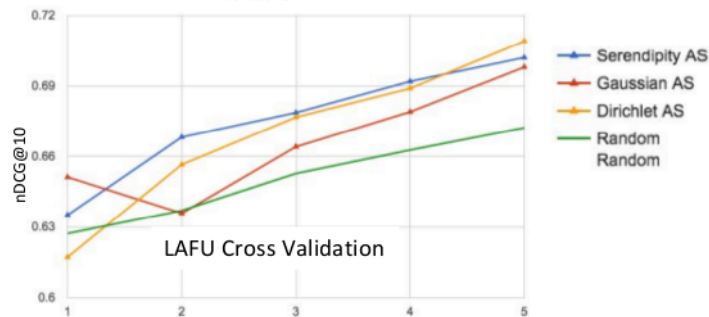
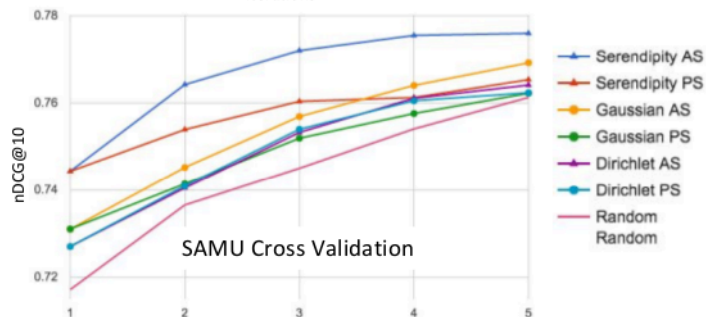
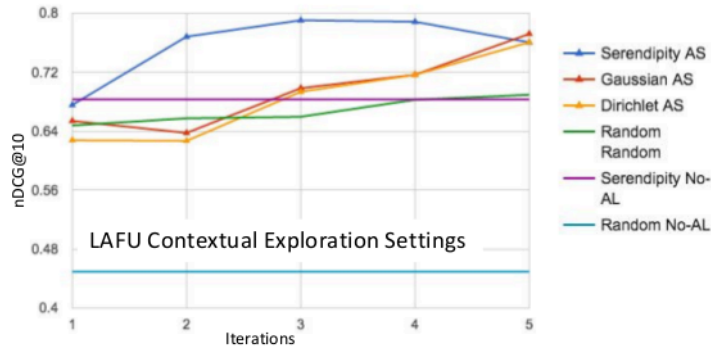
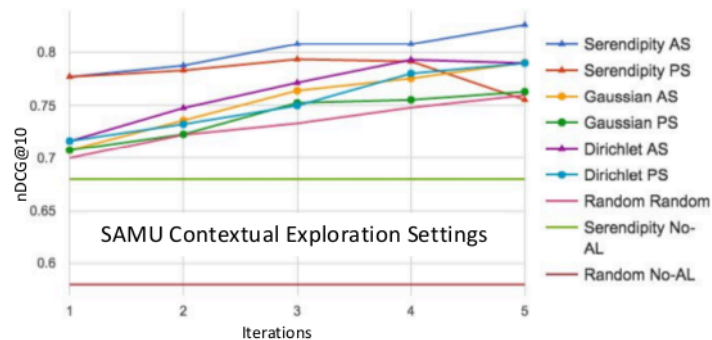
These measure are usually defined in a range [0, 1]:

- 0 => complete disagreement between users
- 1 => users give unanimous rates

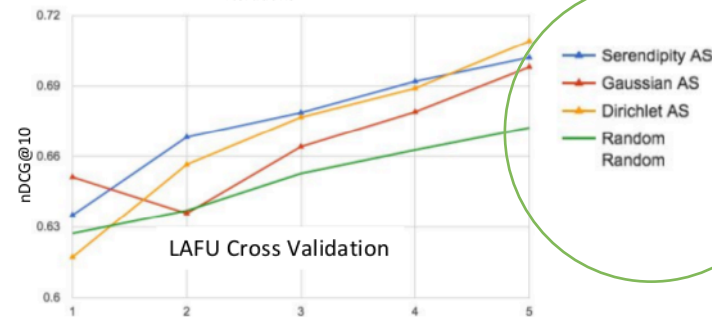
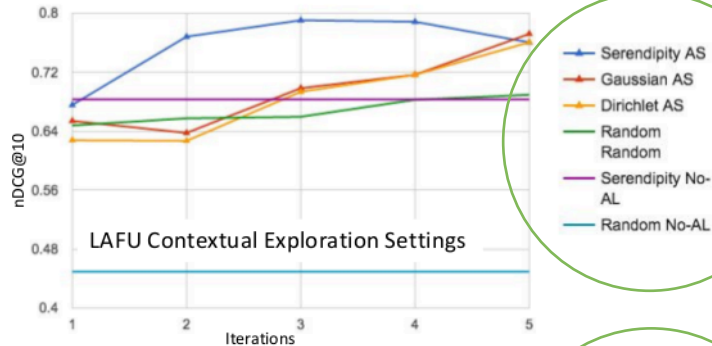
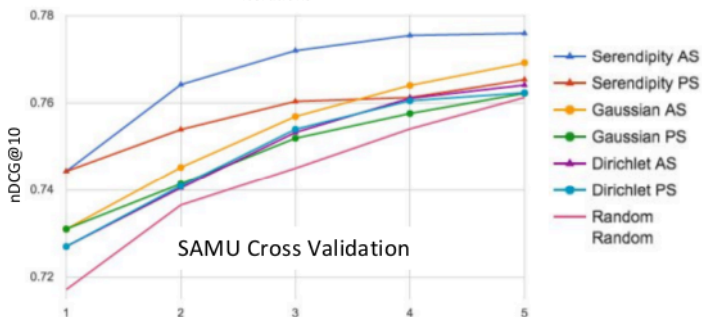
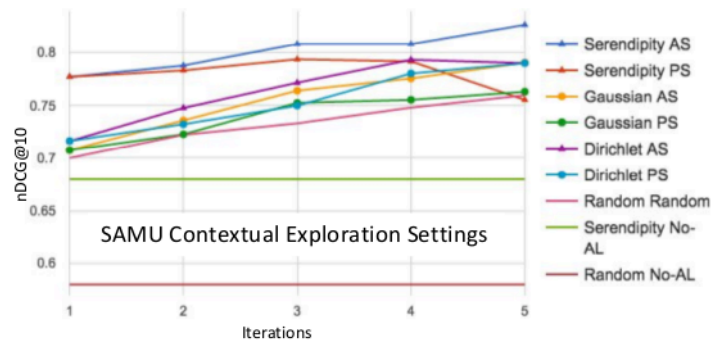
Krippendorff's alpha	0.061
Kendall's W	0.26

Value are far from 1 => hypothesis validated

Results: Performance (nDCG@10)

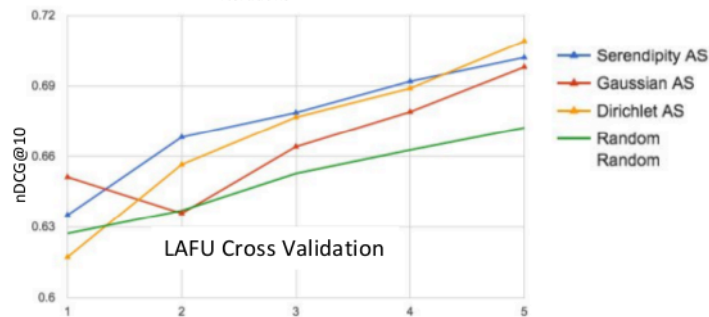
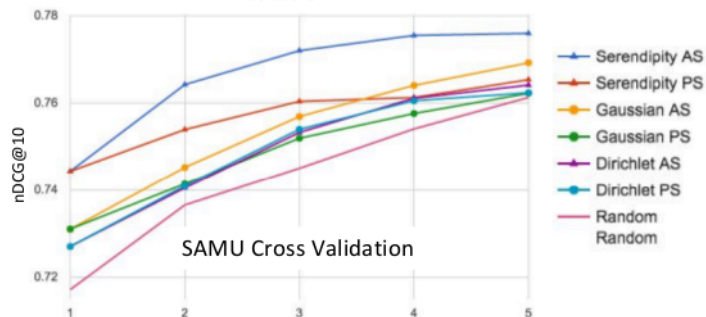
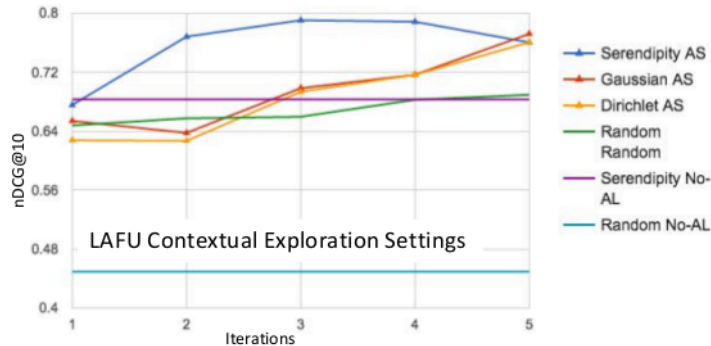
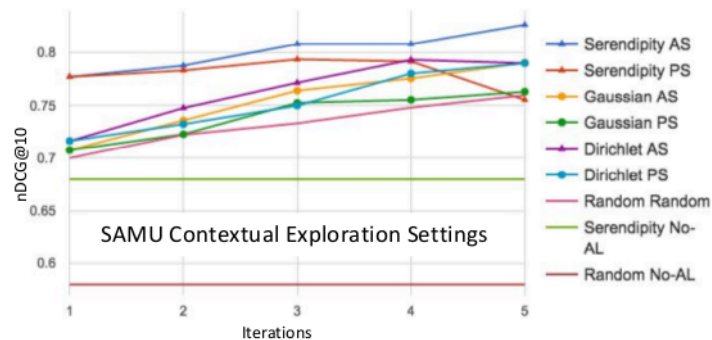


Results: Performance (nDCG@10)

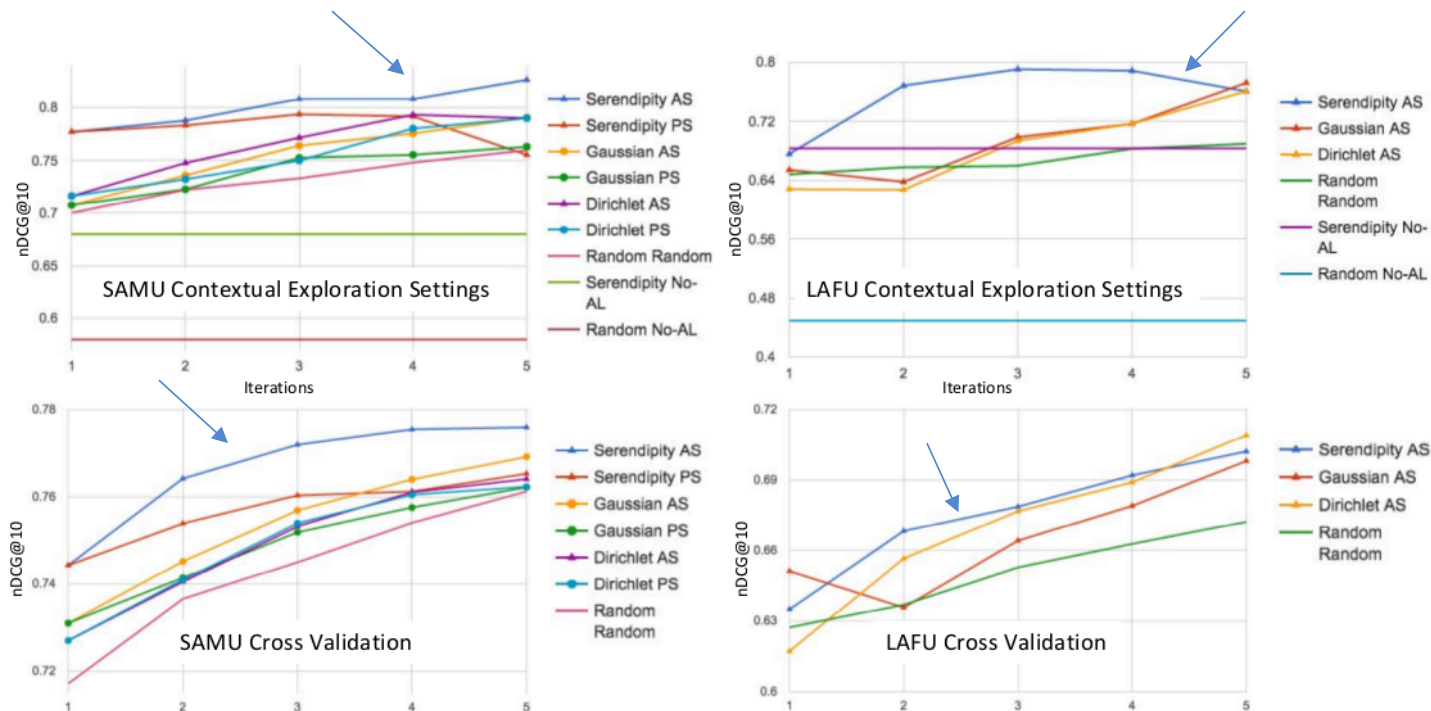


No PS active sampling, no
real time usage possible

Results: Performance (nDCG@10)

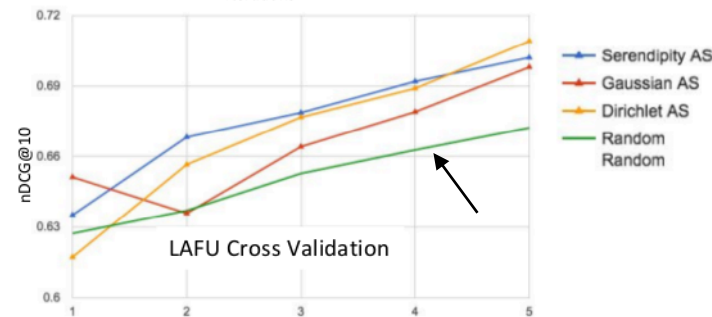
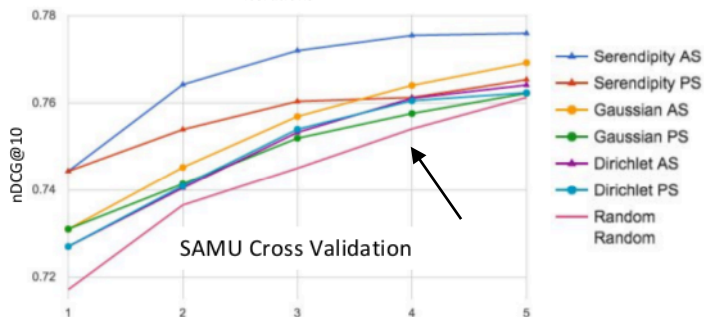
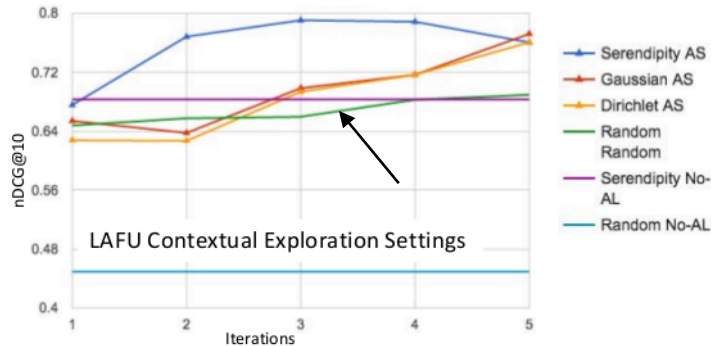
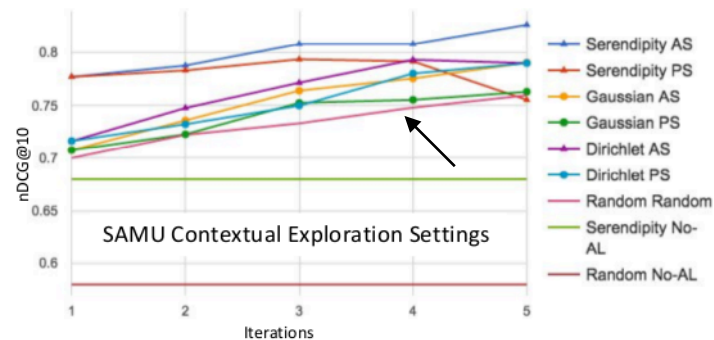


Results: Performance (nDCG@10)



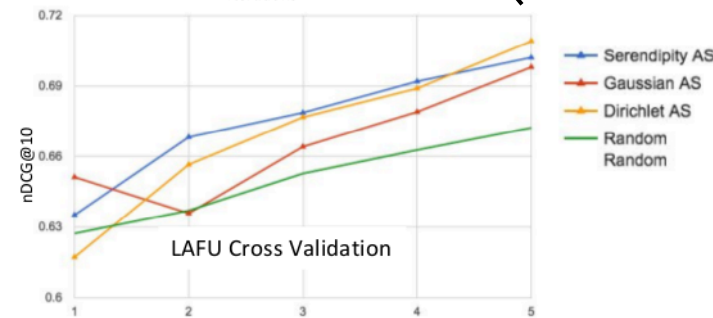
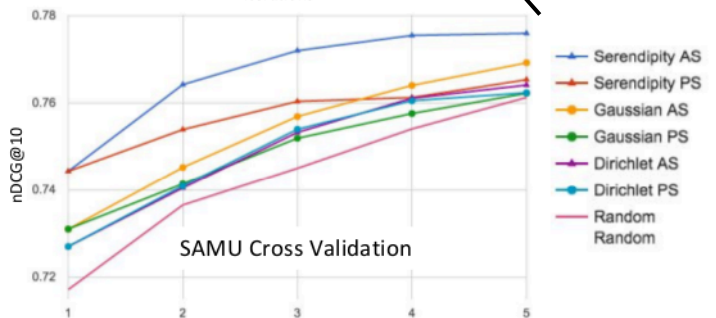
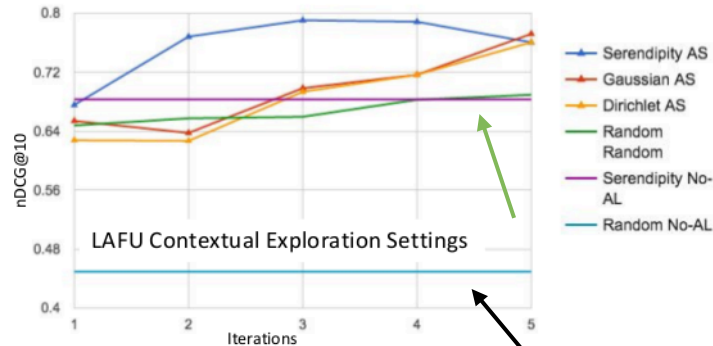
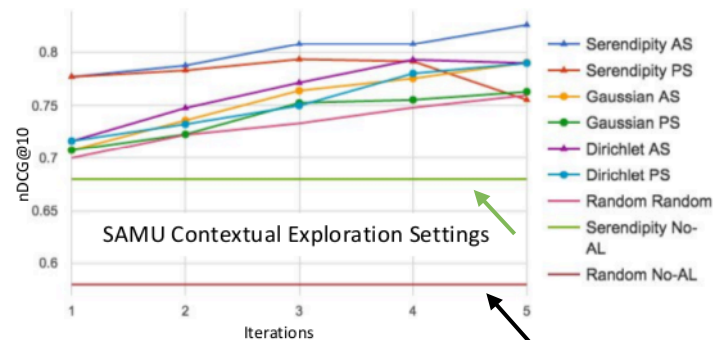
Serendipity AS
(AUC Based Sampling)

Results: Performance (nDCG@10)



Random Random Baseline

Results: Performance (nDCG@10)



Random Baseline
Serendipity Baseline

Ongoing Work on DaCENA User Interface

- 1) New interface design
- 2) Comprehensive evaluation of the application after new interface

The screenshot displays the DaCENA web application interface. On the left, a news article titled "Bernie Sanders: I will work with Hillary Clinton to stop Donald Trump" is shown, with entities like "Vermont", "Clinton", and "Trump" highlighted. The main area features a network graph with nodes representing entities (e.g., Donald Trump, Democratic Party, Hillary Clinton, Vermont, Bernie Sanders) and relationships (e.g., COSA, TIZIO, PAESE). The graph is interactive, with controls for "N° nodes", "Expand node", and "Shrink node". A "MAP" section on the right shows a map of the United States. Below the map, "TOP ASSOCIATIONS" are listed, including "Donald Trump > PARTY > Democratic Party < PARTY < Hillary Clinton" and "Peter Welch < SUCCESSOR < Bernie Sanders > PARTY > Democratic Party". The interface also includes a search bar and a "Link" button.

New Interface Design: Evaluation Methodology



Method

Task Analysis and
thinking aloud with
final interview



Users

11 users age
between 20 and
35



Measures

Success Index,
Net Promoter
Score (NPS)

New Interface Design: Evaluation Results



Success Index: 90%, for all the tasks considered

Measures

NPS: 36, where a value greater than 0 is a good result while a value greater than 50 is an excellent result

Outline

- Contextual Exploration of Knowledge Graphs
- Actively Learning to Rank Semantic Associations
- Experiments
- **Conclusions and Future Work**

Conclusions and Future Work

Conclusions:

1. Quick optimization of personalized ranking function with Active Learning to Rank
2. Active Learning to Rank can be initialized with Serendipity (+performance, +interaction flow)

Future Work:

1. Exploring new algorithms for the active learning to rank
2. Need to better understand how to design user interaction for the ALR model

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